

Random Matrices

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Definition 0.1. Consider a matrix of random entries:

$$\begin{pmatrix} * & \cdots & * \\ \vdots & \ddots & \vdots \\ * & \cdots & * \end{pmatrix}$$

These entries are random variables. Let $X_1, \dots, X_n$ be vectors in $\mathbb{R}^d$, where each $X_i = (X_{i1}, \dots, X_{id})$ are random variables.

Definition 0.2. The variance of a random variable X_{ij} is defined as:

$$\frac{1}{n} \sum_{i=1}^n X_{ii}^2 \longrightarrow \text{Variance}$$

The covariance between two random variables X_{i1} and X_{i2} is defined as:

$$\frac{1}{n} \sum_{i=1}^n X_{i1} X_{i2} \longrightarrow \text{Covariance}$$

Let X be the data matrix:

$$X = \begin{pmatrix} X_{11} & \cdots & X_{1d} \\ \vdots & \ddots & \vdots \\ X_{n1} & \cdots & X_{nd} \end{pmatrix}$$

The sample covariance matrix is given by:

$$\frac{XX^\top}{n}$$

Definition 0.3. The eigenvalues λ of a matrix A are defined by the equation:

$$Av = \lambda v \quad \text{for } v \neq 0$$

The trace of a matrix A is the sum of its eigenvalues:

$$\text{tr}(A) = \sum_{i=1}^n A_{ii} = \sum_{i=1}^n \lambda_i$$

Remark 0.1. The set of eigenvalues of a random matrix is called the empirical spectral distribution:

$$\frac{1}{n} \sum_{i=1}^n \delta_{\lambda_i}$$

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Consider a random matrix $A = \begin{pmatrix} A_{11} & A_{1j} & \cdots \\ \vdots & \vdots & \ddots \\ A_{m1} & A_{mj} & \cdots & A_{mn} \end{pmatrix}$ where the entries are ± 1 with probability $\frac{1}{2}$.

Note that $A_{ji} = A_{ij}$.

Trace properties:

$$\text{tr}(A) = \sum_i A_{ii} = \sum_i \lambda_i,$$

$$\text{tr}(A^2) = \sum_i A_{ii}^2 = \sum_i \lambda_i^2,$$

$$\text{tr}(A^k) = \sum_i A_{ii}^k = \sum_i \lambda_i^k.$$

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